DHAIRYARAJ SINGH SHEKHAWAT

Third-Year Electrical Engineering Student · Power Electronics, Batteries, EV Systems
Pune, India | shekhawatds2006@gmail.com
linkedin.com/in/dhairyaraj-singh-shekhawat-757057327

EDUCATION

Dr. D. Y. Patil Institute of Technology, Pimpri (Savitribai Phule Pune University) **2024 – 2028**
B.E., Electrical Engineering · CGPA 8.84/10 (Sem I–IV) *Pune, India*

Indian Institute of Technology Guwahati **2024 – 2028**
B.Sc. (Hons), Data Science & Artificial Intelligence · online degree, pursued in parallel *Online*

Saraswati Bal Niketan Sr. Sec. School · Class XII: 83.60% *Rajasthan, India*

Coursework: Power Electronics, Electrical Machines, Power Systems, Network Analysis, Analog & Digital Electronics

PROJECTS

Sodium-Ion Cell Recovery Test Bench **Sep 2026 – present**
TE seminar and group project · LTspice · ongoing

- Working on a new way to recover sodium-ion cells after 0 V storage, based on a 12-source literature review; bench build is next.
- Simulated the bench's sensor amplifier in LTspice. The first single-stage LM358 design gave only ×3.6 at 300 kHz, so I redesigned it as two ×10 stages with a faster op-amp: ×83.7 peak gain over 59.6–543 kHz.
- In LTspice, stabilised the charging current source with a 1 nF compensation capacitor: a 10× current step that peaked 87% above target and kept ringing now settles in 52 μs.

Bidirectional SiC Dual-Active-Bridge DC-DC Converter for Vehicle-to-Grid (V2G) **Jul 2026**
Independent simulation study · MATLAB

- Simulated a 3.3 kW DAB (350 V battery, 400 V bus, 50 kHz). My aging-aware model predictive control (MPC) cut overshoot from 17.5% (PI) to 11.3% under load steps, but left a 0.19 V steady-state error and cut simulated aging by only 1.5%.

Step-Down Chopper (DC-DC), Breadboard Build **Oct 2026**
TE power electronics lab, group build · Arduino, IRLB8748 MOSFET

- Built a 24 V step-down chopper: Arduino PWM drives an IRLB8748 logic-level MOSFET (220 Ω gate resistor, 10 kΩ pull-down), with an SB560 Schottky freewheel diode and a 470 μF input capacitor; a potentiometer sets the duty ratio.
- Measured output voltage against duty ratio (first test: 8.1 V in, 4.03 V out at D ≈ 0.5); fixed a missing common ground between the Arduino and the power stage.

Electric Go-Kart Power Path & Safety Loop **Aug – Oct 2026**
Team The Pegasus (K2E05), electrical sub-team · FKDC Season 10, EV category · design stage

- Made a simple Simulink model of the kart's 72 V power path and safety logic: two 20S14P Li-ion packs in parallel (84 Ah, 6.05 kWh) feeding a 6 kW PMSM.
- Designed the 12 V safety loop: shutdown circuit, two isolation relays, brake over-travel switch and ready-to-drive buzzer.

Off-Grid Solar Sign Planner **Oct 2026**
InUnity Entrepreneurial Mindset Program, Team Illuminators · selected for the proof-of-concept round

- Built a sizing tool that runs the sign's LiFePO4 battery night by night over 10 years of NASA POWER data; it showed a bigger panel with smart dimming beats a wind turbine in the city, with zero dark nights in 10 years.
- Simulated a 20 kHz buck charger with P&O MPPT in Simulink (6 × 540 W PV array into a 25.6 V LiFePO4 battery): it held ≈3.03 kW at 1000 W/m² and found the new maximum after irradiance steps.

MineAdapt-EC — Solar Electrocoagulation & Water-Quality Monitor **Sep 2026**
Smart India Hackathon 2026 · SIH26040, Govt. of Jharkhand · Team H2ohmies, research & software

- Sized the iron dose for a solar-powered reactor for mine water with Faraday's law (41.7 mg/L at 1.2 A, 0.5 L/min). In our beaker bench test (22.6 V, 2.57 A), floc formed as expected.
- Wrote and flashed ESP32-C6 firmware for pH, turbidity, TDS and temperature with a live Wi-Fi dashboard; probe tests pending.

Lab simulations: square-wave inverter (THD 47.3%, theory 48.3%), SPWM inverter, single-phase full converter · **Also:** grid fault classifier (PyTorch)

SKILLS

Power & embedded	MATLAB/Simulink, LTspice, Arduino and ESP32 (basic C/C++), breadboard builds, multimeter
Programming & data	Python (pandas, NumPy, Matplotlib, scikit-learn, PyTorch), MATLAB scripting
AI-assisted coding	Claude Code for project code: ESP32 firmware, Python tools, dashboards
Languages	English, Hindi

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon 2026: cleared the college-level round with MineAdapt-EC; national result awaited.
- ML Fundamentals (DA 261) and Time Series Analysis (DA 210), IIT Guwahati, May 2026 · Python for Data Science, NPTEL **Elite**